

FALL 2026

**NOW
FULLY IN
PYTHON**



CIC 1203



9 UNITS (15-322)
12 UNITS (15-622)

DIGITAL AUDIO

$x(t) \rightarrow \text{sample} \rightarrow 3\text{-bit quantize}$ 8 amplitude levels

— continuous signal ○ samples ■ quantized values

SYNTHESIS

ADDITIVE: $x(t) = \sum_{k=1}^K a_k \sin(2\pi f_k t)$

$\xrightarrow{\text{SIFT}}$

$k=1, a_k=1$
 $k=2, a_k=0.53$
 $k=3, a_k=0.3$

FM: $y(t) = \sin(2\pi f_c t + I \sin(2\pi f_m t))$

modulator $\sin(2\pi f_m t)$ $\times$ $2\pi f_c t$ $+$ $\sin(\cdot)$ $y(t)$

FOURIER & DSP

TIME DOMAIN

DFT MAGNITUDE

3 7 11

$y[n] = (x * h)[n]$

33-TAP WINDOWED-SINC LOW-PASS FIR

$|H(f)|$

1

0

0

1 x Nyquist

EFFECTS & SPATIALIZATION

STEREO TAPPED DELAY LINE

$D = \text{one delay interval}$

Diagram illustrating a Stereo Tapped Delay Line. The input signal $x[n]$ is processed through a series of delay blocks (z^{-D}) to produce delayed versions of the signal. The output of the first delay block is $x[0.55]$, the output of the second is $x[0.25]$, and the output of the third is $x[0.00]$. These delayed signals are then combined with the original input signal $x[n]$ to produce the final stereo signals $y_L[n]$ and $y_R[n]$.

For the Left channel ($y_L[n]$):

$$y_L[n] = x[n] + 0.55x[n-D] + 0.25x[n-2D]$$

For the Right channel ($y_R[n]$):

$$y_R[n] = x[n] + 0.20x[n-D] + 0.35x[n-2D]$$

ALGORITHMIC COMPOSITION

MARKOV TRANSITION MATRIX

Columns = next state

rows = current	C4	E4	G4
C4	0.10	0.60	0.30
E4	0.30	0.20	0.50
G4	0.60	0.30	0.10

SEQUENCE FROM SEED 32

CEECEGCEEEEGEGGEC

16 quarter-note steps

REAL-TIME AUDIO & MUSIC AI

REAL-TIME FRAME PIPELINE

frame N=512 · hop H=256

```
graph LR; RB[RING BUFFER] --> HW[HANN WINDOW]; HW --> STFT[|STFT|]; STFT --> MF[MIR FEATURES]; MF --> M[MODEL]; M --> Out[ ];
```

audio callback reduces leakage time → frequency spectral descriptors prediction / control



PREREQUISITE:
15-122 OR 15-112



**MUSIC EXPERIENCE
NOT REQUIRED**



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